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A Metal-Poor Tidal Disruption Engine: The Warm Molecular Equilibrium and Its Application to Little Red Dots

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ABSTRACT

The unbound debris of tidal disruption events can sustain a self-regulating equilibrium in galactic nuclei: momentum injected into surrounding molecular clouds compresses the nuclear star cluster until stellar collisions cap the disruption rate at $\Gamma_{\rm eq} \simeq 1.4 \times 10^{-2}\, M_{\rm h,6}^{-0.84}\,{\rm yr}^{-1}$ (Guillochon & Loeb 2026). That solution relies on CO and dust to hold the clouds at 10 K and does not apply to the metal-poor nuclei in which little red dots (LRDs) form. We re-derive it with H_2 rotational cooling in local thermodynamic equilibrium and find that the engine survives on a warm branch. The collision cap and the belt radius do not involve cooling, so $\Gamma_{\rm eq}$, $r_{\rm b} \simeq 4.9\, M_{\rm h,6}$ pc and $n_{\rm MC} \simeq 1.9 \times 10^5\, M_{\rm h,6}^{-2}\,{\rm cm}^{-3}$ carry over unchanged. The keV emission of the radiative remnants heats the H_2-cooled clouds volumetrically. Their temperature is bracketed by the requirements that the clouds fit within the disk and remain supersonically turbulent, $740\, M_{\rm h,6}^{-0.40} \lesssim T_{\rm MC} \lesssim 1100\, M_{\rm h,6}^{-0.36}$ K. In this window the cloud radii, gas mass ($\sim 3 \times 10^5\, M_\odot$) and mass floor ($\sim 10^6\, M_\odot$) match the metal-rich solution to within factors of two, but the observables differ sharply. The extinction falls to $A_V \simeq 27\, Z'$. The belt dust is heated only by the luminosity averaged over the duty cycle, so it reradiates ~ 1 per cent of the flare luminosity at ~ 60 K, consistent with the dust deficit of LRDs. The injected energy emerges as H_2 lines with $L_{\rm H_2} \simeq 8 \times 10^{40}\, M_{\rm h,6}^{-0.5}\,{\rm erg\,s}^{-1}$. The engine operates for $10^6 \lesssim M_{\rm h} \lesssim (1.5\text{–}5) \times 10^6\, M_\odot$ and $10^{-3} \lesssim Z' \lesssim 0.3$, a range that contains the measured metallicities of LRD hosts.

Keywords: Tidal disruption (1696) — Active galactic nuclei (16) — Supermassive black holes (1663) — Molecular clouds (1072) — High-redshift galaxies (734)

1. INTRODUCTION

JWST has uncovered compact, red, broad-lined sources at $z \sim 4$–9, known as little red dots (LRDs; Matthee et al. 2024; Greene et al. 2024; Harikane et al. 2023; Kocevski et al. 2025). They are abundant, with number densities of order 10^{-5}–10^{-4} cMpc^{-3} (Matthee et al. 2024; Kokorev et al. 2024; Akins et al. 2025). They are also compact, and their spectral energy distributions are puzzling: they show a marked deficit of both hot and cold dust emission (Setton et al. 2025; Chen et al. 2025; Casey et al. 2025) alongside reddening that may arise from dense gas rather than dust (Inayoshi & Maiolino 2025; Naidu et al. 2026; de Graaff et al. 2026). Their hosts are metal-poor, with direct-method oxygen abundances of $Z' \simeq 0.08$ and a spread of ~ 0.6 dex (Nikopoulos et al. 2026). Among the interpretations proposed for them is that they are powered by the recurrent tidal disruption of stars in nuclei that have recently built a black hole by runaway stellar collisions (Bellovary 2025; Pacucci et al. 2025; Kritos & Silk 2026a). The radio-quiet character of the population is consistent with this (Perger et al. 2025), and the residual stellar cores left behind by seed formation are dense enough to sustain high disruption rates (Pacucci et al. 2025).

What that picture lacks is a mechanism that sets the disruption rate. Guillochon & Loeb (2026, hereafter GL26) supply one. In that model the unbound debris of each disruption drives a supernova-like remnant into the molecular clouds surrounding the nucleus. The momentum keeps the clouds turbulent and dense, the dense clouds shorten the relaxation time of the nuclear star cluster, and the compressed cluster disrupts stars faster still. The runaway is arrested when stars begin to collide, and the steady state has $\Gamma_{\rm eq} \simeq 1.4 \times 10^{-2}\, M_{\rm h,6}^{-0.84}\,{\rm yr}^{-1}$, where $M_{\rm h,6} \equiv M_{\rm h}/10^6\, M_\odot$. GL26 noted (§7.2) that the equilibrium rates and stellar densities coincide with those inferred for LRD cores, but also identified two obstacles to applying the model there.

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The first is obscuration: the metal-rich engine is buried under $A_V \simeq 50$, whereas LRDs lack dust emission. The second is metallicity: in the condition of energy balance of GL26 the clouds cool through CO lines and are held at 10 K by the balance of cosmic-ray heating and molecular cooling, and neither CO cooling nor that balance is available in gas with $Z \lesssim 0.1\,Z_\odot$. As GL26 put it, the condition "would have to be rederived with molecular hydrogen or atomic cooling."

This paper carries out that re-derivation. We keep every element of the GL26 framework and change only the microphysics of the cooling. The elements that we keep are the Bahcall–Wolf cusp, the tidally marginal cloud belt, the covering, energy, collision and compression conditions, and the monomial method of solution. The parts that change are the volumetric cooling rate, the sound speed of the clouds, and the radiative time of the unbound debris remnants, each of which depends on composition. The structure of the GL26 solution turns out to isolate the effect of metallicity cleanly. Of the four unknowns describing the equilibrium, two are fixed by conditions that do not involve cooling at all, and the two that remain inherit the composition dependence in a way that can be written as exact power laws in the cloud temperature, the H_2 abundance, and the metallicity.

The paper is organized as follows. Section 2 summarizes the GL26 equilibrium and identifies where composition enters. Section 3 describes the cooling of metal-poor gas at the densities of interest and argues that the cold, CO-cooled branch of the equilibrium cannot exist there. Section 4 solves for the warm, H_2-cooled branch and shows that two requirements of internal consistency bracket the cloud temperature, which in GL26 was an external input. Section 5 draws out the observable consequences for LRDs, Section 6 discusses the limitations of the treatment, and Section 7 summarizes. Throughout, normalized quantities are indicated by a subscript, so that $T_3 \equiv T_{\rm MC}/10^3$ K, $Z' \equiv Z/Z_\odot$, and $x_{\rm H_2}$ is the fraction of hydrogen nuclei bound in H_2.

2. THE ENGINE AND WHERE COMPOSITION ENTERS

We follow the notation of GL26, to which we refer the reader for the derivations; here we restate only what is needed to see where the cooling function appears.

The nucleus is a relaxed Bahcall–Wolf cusp, $n_* \propto r^{-7/4}$, containing a stellar mass $2M_{\rm h}$ within the sphere of influence $a_{\rm h} = 2GM_{\rm h}/\sigma^2$, with mean stellar mass $m_* = 0.5\,M_\odot$ and radius $R_* = 0.46\,R_\odot$ (Tout et al. 1996). The stars occupy a fraction f_* of 4π steradians. Molecular clouds orbit in a belt of radius $r_{\rm b}$ and aspect ratio $f_\Omega = H/r_{\rm b}$, where they are marginally tidally bound,

$$\rho_{\rm MC} = \frac{9M_{\rm h}}{4\pi r_{\rm b}^3}, \tag{1}$$

and supported against self-gravity by supersonic turbulence,

$$\mathcal{M}^2 c_{\rm s}^2 = \frac{4\pi}{5} G\rho_{\rm MC} R_{\rm MC}^2, \tag{2}$$

with $\sigma_{\rm cl} = \mathcal{M}c_{\rm s}$ the turbulent velocity and $\tau_{\rm c} = 2R_{\rm MC}/\sigma_{\rm cl}$ the crossing time on which the turbulence decays. The number of clouds needed to pave the belt is $N_{\rm MC} = 4f_\Omega r_{\rm b}^2/R_{\rm MC}^2$, and the stars are fed from the disk so that the debris is beamed back into it ($b = 1$ in the notation of GL26).

Four conditions close the system. The first (GL26 eq. 23) requires every patch of the belt to be struck by an unbound debris remnant (UDR) at least once per crossing time,

$$\Gamma_{\rm TDE}\, f_{\rm UDR}\, \tau_{\rm c}\, \beta = 1, \qquad f_{\rm UDR} = \frac{R_{\rm UDR}^2}{4r_{\rm b}^2}, \tag{3}$$

with $\beta = b/f_\Omega$ and $R_{\rm UDR}$ the radius at which a remnant of energy $E_{\rm UDR} = 1.8\times10^{50}\,M_{\rm h,6}^{1/3}$ erg (Guillochon et al. 2016) becomes radiative in gas of density $\rho_{\rm MC}$ (Sedov 1946; Taylor 1950; Blondin et al. 1998). The second (GL26 eq. 27) requires the clouds to radiate the energy they receive,

$$\Gamma_{\rm TDE}\, E_{\rm UDR}\, b = L_{\rm MC} = N_{\rm MC} V_{\rm MC}\, \Lambda_{\rm vol}, \tag{4}$$

where $V_{\rm MC} = 4\pi R_{\rm MC}^3/3$ and $\Lambda_{\rm vol}$ is the volumetric cooling rate of the cloud gas. The third (GL26 eq. 29) caps the disruption rate at the rate at which stars collide before reaching the loss cone,

$$\Gamma_{\rm coll}/f_* = \Gamma_{\rm TDE}, \tag{5}$$

where $\Gamma_{\rm coll}$ is the gravitationally focused collision rate integrated over the cusp (GL26 eq. 30) and $\Gamma_{\rm TDE}$ is the bridged Cohn–Kulsrud flux into the loss cone (Cohn & Kulsrud 1978; Merritt 2013; Vasiliev & Merritt 2013; Stone & Metzger 2016). The fourth (GL26 eq. 32) is an inequality rather than a closure: the clouds must dominate the relaxation of the cusp, $t_{\rm relax,MC} \le t_{\rm relax,*}$, which sets a lower bound on $M_{\rm h}$.

Composition enters in three places. The cooling rate $\Lambda_{\rm vol}$ in equation (4) is the most obvious; GL26 take $\Lambda_{\rm vol} = \Lambda_0 n_{\rm MC}^2$ with $\Lambda_0 = 1.3\times10^{-27}$ erg cm^3 s^{-1}, appropriate to CO at 10 K (Goldsmith & Langer 1978; Neufeld et al. 1995). The sound speed $c_{\rm s}$ in equation (2) depends on the cloud temperature and mean molecular weight; GL26 take $c_{\rm s} = 0.19\,{\rm km\,s^{-1}}$ for 10 K gas at $\mu = 2.33$. The third is the radiative time of the

remnant, $\tau_{\rm rad}$, which sets $R_{\rm UDR}$ in equation (3). It depends on the cooling of the hot, shocked phase, and GL26 took it from numerical calculations at solar composition (Blondin et al. 1998). No other part of the system depends on metallicity. In particular, the collision condition (5) involves only the cusp, so the equilibrium dispersion and disruption rate,

$$\sigma_{\rm eq} = 2.4\times10^{2}\,M_{\rm h,6}^{0.03}\,{\rm km\,s^{-1}},$$
$$\Gamma_{\rm eq} \simeq 1.4\times10^{-2}\left(\frac{f_*}{0.22}\right)^{2.4} M_{\rm h,6}^{-0.84}\,{\rm yr^{-1}}, \qquad (6)$$

are the same for any composition, as are $a_{\rm h} = 0.15\,M_{\rm h,6}^{0.94}$ pc and the central density $\rho_0 = 6\times10^{7}\,M_{\rm h,6}^{-1.8}\,M_\odot\,{\rm pc^{-3}}$ at $f_* = 0.22$. Metallicity affects the gas, not the stars.

3. COOLING OF METAL-POOR GAS IN THE BELT

3.1. *Why the cold branch does not exist at low metallicity*

In the GL26 solution the cloud temperature is not an output of the engine but an external thermostat: the clouds sit at ~ 10 K because CO and dust cooling are so steep below H recombination that the gas rapidly settles where cooling balances cosmic-ray heating (Krumholz 2015). The engine then adjusts the cloud size, count, and mass so that the CO luminosity of the belt matches the UDR power.

Both halves of that thermostat scale with metallicity. The CO abundance, and hence Λ_0, is proportional to Z' for $Z' \lesssim 1$, and the dust that thermally couples to the gas at the highest densities is likewise diluted. If we reduce Λ_0 by Z' in equation (4) and hold the temperature at 10 K and the other conditions fixed, the required cloud radius grows as $R_{\rm MC} \propto Z'^{-1}$. For $Z' \lesssim 0.7$ the clouds outgrow the disk half-thickness $H \simeq 1$ pc that is supposed to contain them, and at $Z' = 0.1$ they have radii of 7 pc. The solution therefore fails the geometric requirement that GL26 use to bound f_* from below. Holding the gas at 10 K is in any case not permitted. The thermostat is gone with the CO and dust that set it. At the redshifts of LRDs, moreover, the cosmic microwave background alone imposes a floor of $T_{\rm CMB} = 16$–22 K at $z = 5$–7, so even the metal-rich values of GL26 that we compare against should be read there as describing a ~ 20 K state rather than a 10 K state. The remaining low-temperature coolants in gas with $Z' \lesssim 0.1$ are the [C II] 158 μm and [O I] 63 μm fine-structure lines, whose efficiency also scales with Z', and below a critical metallicity of order $10^{-3.5}\,Z_\odot$ they cease to be competitive with H_2 (Bromm & Loeb 2003; Omukai 2000; Santoro & Shull 2006; Glover & Jappsen 2007). This argument does not exclude an intermediate state at $Z' \sim 0.3$–0.7, in which CO and fine-structure lines cool clouds that do fit within the disk. The lines would do so at a few tens to a few hundred K, where they radiate far more per nucleus than at 10 K. We return to that regime in Section 5.5. What it does establish is that there is no metal-poor analogue of the 10 K state that the GL26 engine can occupy.

3.2. H_2 *cooling at the belt density*

What metal-poor gas does have is molecular hydrogen. At the densities that the equilibrium will return, $n_{\rm MC} \sim 10^5$ cm^{-3}, the rotational levels of H_2 are thermalized. The critical density for the pure rotational cooling is of order 10^3–10^4 cm^{-3} at $T \sim 10^3$ K for collisions with atomic hydrogen (Hollenbach & McKee 1979; Galli & Palla 1998; Glover & Abel 2008), and up to $\sim 10^5$ cm^{-3} when H_2 itself is the collider (Le Bourlot et al. 1999; Coppola et al. 2019). The cooling is therefore in the local thermodynamic equilibrium (LTE) regime and scales with the first power of density rather than the second. We adopt the LTE rotational cooling rate per molecule of Hollenbach & McKee (1979), which over the range $200 \lesssim T \lesssim 2000$ K is well represented by the single power law

$$L_{\rm LTE}(T) \simeq 9.5\times10^{-22}\,T_3^{3.76}\ {\rm erg\,s^{-1}}\ {\rm per\ H_2}, \qquad (7)$$

where we have dropped the correction factors that enter the full fit; they are 0.9 at 10^3 K and 0.65 at 2×10^3 K. The volumetric cooling rate of a cloud with hydrogen-nucleus density $n_{\rm MC}$ and molecular fraction $x_{\rm H_2}$ is then

$$\Lambda_{\rm vol} = x_{\rm H_2}\,\frac{n_{\rm MC}}{2}\,L_{\rm LTE}(T_{\rm MC}). \qquad (8)$$

We take a primordial composition for the belt gas, with hydrogen mass fraction $X = 0.76$, so that $n_{\rm MC} = \rho_{\rm MC}/1.32\,m_{\rm p}$ and, for fully molecular gas, $\mu = 2.27$ and

$$c_{\rm s} = 1.9\,T_3^{1/2}\,{\rm km\,s^{-1}}. \qquad (9)$$

Three properties of equation (7) shape everything that follows. It is steep, $\propto T^{3.76}$, because the lowest excited rotational level lies at $E/k = 510$ K, so the cooling collapses below ~ 200 K (Bromm et al. 2002; Abel et al. 2002). It is linear in density, unlike the n^2 dependence of the sub-thermal CO cooling used by GL26, which alters the monomial structure of the solution. And it carries no external thermostat: nothing in metal-free gas fixes the temperature of dense molecular gas the way cosmic rays and CO fix it at 10 K. The last point means that $T_{\rm MC}$ cannot be imposed. It must emerge from the equilibrium itself, and we show how in Section 4.2.

Vibrational cooling adds to equation (7) at $T \gtrsim 10^3$ K, but its critical density is of order 10^6–10^7 cm^{-3}, so at $n_{\rm MC} \sim 10^5$ cm^{-3} it is sub-thermal and contributes at the tens of per cent level. HD cooling is important only below ~ 200 K (Galli & Palla 1998; Glover & Abel 2008) and does not enter at the temperatures we find. Both effects are absorbed, along with any uncertainty in the normalization of the fit, into $x_{\rm H_2}$, which enters the solution only through the product $x_{\rm H_2} L_{\rm LTE}$.

Metal fine-structure lines are not negligible at the top of the metallicity range we consider. At 10^3 K and $n_{\rm MC} \sim 2\times10^5$ cm^{-3} the [O I] 63 and 145 μm lines radiate $\sim 3\times10^{-19}$ erg s^{-1} per oxygen atom, even though they are sub-thermal by a factor of 2–3 at this density. For $n_{\rm O}/n_{\rm H} = 4.9\times10^{-4} Z'$ this is $\sim 1.5\times10^{-22}\, Z'$ erg s^{-1} per hydrogen nucleus, and [C II] and [Si II] add less (Santoro & Shull 2006; Glover & Jappsen 2007; Meijerink & Spaans 2005). This is $\sim 0.5\, Z'$ of the H_2 rate of equation (7) at the same temperature, so below 15 per cent for $Z' \leq 0.3$ and comparable to H_2 only as $Z' \to 1$. CO would contribute a similar fraction if it survived, but in gas heated by X-rays (Section 4.2) carbon is kept atomic or singly ionized (Maloney et al. 1996). We treat the metal lines as part of the uncertainty of tens of per cent that is absorbed into $x_{\rm H_2}$, and use them to set the upper edge of the validity range in Section 5.5.

3.3. *The radiative phase of the remnants*

The radius $R_{\rm UDR}$ at which a debris remnant deposits its momentum is set by when its shocked interior becomes radiative. At the belt density this happens early and fast: the shell is still moving at $\sim 10^3$ km s^{-1} when $\tau_{\rm rad}$ is reached (eqs. 14–15 below). The post-shock temperature is therefore $\sim 10^7$ K, and the remnant radiates as ∼keV bremsstrahlung rather than in the 10^5–10^6 K lines of a classical interstellar remnant. GL26 took $\tau_{\rm rad} = 130\, E_{50}^{4/17} n_{\rm MC,4}^{-9/17}$ yr from Blondin et al. (1998), computed at solar abundances. At 10^7 K the metals matter less than they would at lower temperature: the metal-free cooling function of Sutherland & Dopita (1993) lies a factor of 2–3 below the solar one there, compared with the order of magnitude that separates the two at 10^5–10^6 K. A Sedov–Taylor remnant becomes radiative after $\tau_{\rm rad} \propto \Lambda_{\rm hot}^{-5/14}$ for the shape of the cooling function assumed by Cioffi et al. (1988). We adopt that scaling, which is an adequate approximation for so weak a dependence, and write

$$\tau_{\rm rad} = 130\, E_{50}^{4/17}\, n_{\rm MC,4}^{-9/17}\, \zeta^{-5/14}\ {\rm yr}, \qquad \zeta \equiv \frac{\Lambda_{\rm hot}}{\Lambda_{\rm hot,\odot}}, \tag{10}$$

with $\zeta \simeq 0.3$–0.5 for $Z' \lesssim 10^{-2}$ and $\zeta \to 1$ as $Z' \to 1$; simulated remnants show a comparably weak dependence on metallicity (Thornton et al. 1998). The dependence is weak, $R_{\rm UDR} \propto \zeta^{-1/7}$ at fixed density, and the terminal momentum of the remnant is insensitive to it in any case (Kim & Ostriker 2015). We carry ζ explicitly, quote its exponents, and evaluate numbers at $\zeta = 1$. At $\zeta = 0.4$ the belt density rises by 50 per cent, the belt radius falls by 13 per cent and the extinction rises by 30 per cent (eqs. 11, 12 and 33). These are the sizes of the corrections that a reader should attach to the metal-poor numbers.

4. THE WARM EQUILIBRIUM

4.1. *Structure of the solution*

With the loss-cone and collision rates evaluated exactly as in GL26, equation (5) fixes $\sigma_{\rm eq}(M_{\rm h})$ and $\Gamma_{\rm eq}(M_{\rm h})$, equation (6), independently of the gas. The belt is then described by the three remaining unknowns $(r_{\rm b}, R_{\rm MC}, \mathcal{M})$ and, unlike in GL26, a fourth, $T_{\rm MC}$. Equations (2), (3) and (4) are each monomial in these unknowns and in $(M_{\rm h,6}, T_3, x_{\rm H_2}, \zeta)$, so the system reduces to a linear solve in logarithms, exactly as in GL26, and we obtain closed-form power laws.[2] With three equations for four unknowns, the belt is a one-parameter family in $T_{\rm MC}$. The physics that selects a value of $T_{\rm MC}$ is the subject of Section 4.2.

The solution has a transparent structure that is worth stating before quoting numbers. Combining equations (1) and (2), the crossing time is $\tau_{\rm c} = 2R_{\rm MC}/\sigma_{\rm cl} = 2(4\pi G\rho_{\rm MC}/5)^{-1/2}$, which depends on $r_{\rm b}$ alone and not on $R_{\rm MC}$ or $T_{\rm MC}$; it is a fixed fraction of the orbital period at $r_{\rm b}$. Since $R_{\rm UDR}$ also depends only on $\rho_{\rm MC}(r_{\rm b})$, the covering condition (3) is an equation for $r_{\rm b}$ alone. The belt radius, cloud density and crossing time are therefore set entirely by the disruption rate and the remnant physics, and inherit no dependence on the cooling function or the temperature:

$$r_{\rm b} = 4.9\, M_{\rm h,6}^{1.00}\, \zeta^{0.15}\ {\rm pc}, \tag{11}$$

$$n_{\rm MC} = 1.9\times10^5\, M_{\rm h,6}^{-2.00}\, \zeta^{-0.44}\ {\rm cm^{-3}}, \tag{12}$$

$$\tau_{\rm c} = 2.4\times10^5\, M_{\rm h,6}^{1.00}\, \zeta^{0.22}\ {\rm yr}, \tag{13}$$

$$R_{\rm UDR} = 0.078\, M_{\rm h,6}^{0.92}\, \zeta^{0.04}\ {\rm pc}, \tag{14}$$

$$\tau_{\rm rad} = 31\, M_{\rm h,6}^{1.14}\, \zeta^{-0.13}\ {\rm yr}. \tag{15}$$

These are the GL26 values to within the small ζ corrections. The energy condition (4) then fixes the cloud radius at given $T_{\rm MC}$, since $L_{\rm MC} = N_{\rm MC} V_{\rm MC} \Lambda_{\rm vol} \propto f_\Omega r_{\rm b}^2 R_{\rm MC}\, n_{\rm MC} x_{\rm H_2} T_3^{3.76}$, and the virial condition (2) fixes the Mach number from $R_{\rm MC}$ and $c_{\rm s}$. For $f_* = f_\Omega = 0.22$,

[2] The scripts that perform the solve, built on the public GL26 code, are released with this paper.

the midpoint of the window found by GL26,

$$R_{\rm MC} = 0.35\, M_{\rm h,6}^{-0.51}\, T_3^{-3.76}\, x_{\rm H_2}^{-1}\, \zeta^{0.15}\ {\rm pc}, \tag{16}$$

$$\mathcal{M} = 1.5\, M_{\rm h,6}^{-1.51}\, T_3^{-4.26}\, x_{\rm H_2}^{-1}\, \zeta^{-0.07}, \tag{17}$$

$$\sigma_{\rm cl} = 2.9\, M_{\rm h,6}^{-1.51}\, T_3^{-3.76}\, x_{\rm H_2}^{-1}\, \zeta^{-0.07}\, {\rm km\,s^{-1}}, \tag{18}$$

$$M_{\rm MC} = 1.1\times10^{3}\, M_{\rm h,6}^{-3.53}\, T_3^{-11.3}\, x_{\rm H_2}^{-3}\ M_\odot, \tag{19}$$

$$N_{\rm MC} = 1.7\times10^{2}\, M_{\rm h,6}^{3.02}\, T_3^{7.52}\, x_{\rm H_2}^{2}, \tag{20}$$

$$M_{\rm disk} = 1.9\times10^{5}\, M_{\rm h,6}^{-0.51}\, T_3^{-3.76}\, x_{\rm H_2}^{-1}\ M_\odot, \tag{21}$$

$$L_{\rm H_2} = \Gamma_{\rm eq} E_{\rm UDR} = 8.1\times10^{40}\, M_{\rm h,6}^{-0.51}\ {\rm erg\,s^{-1}}. \tag{22}$$

Four further quantities follow: the ambient density in pressure balance with the clouds, the accretion luminosity fed by Bondi inflow for a Bondi efficiency $f_{\rm B} = 10^{-3} f_{\rm B,-3}$, the hydrogen column through the belt, and the ratio of disruption-fed to ambient accretion,

$$n_{\rm bg} = 1.8\times10^{3}\, M_{\rm h,6}^{-5.0}\, T_3^{-7.5}\, x_{\rm H_2}^{-2}\, \zeta^{-0.44}\ {\rm cm^{-3}}, \tag{23}$$

$$L_{\rm AGN}/L_{\rm Edd} = 1.2\times10^{-2}\, f_{\rm B,-3}\, M_{\rm h,6}^{-4.0}\, T_3^{-7.5}\, x_{\rm H_2}^{-2}\, \zeta^{-0.22}, \tag{24}$$

$$N_{\rm H} = 2.7\times10^{22}\, M_{\rm h,6}^{-4.0}\, T_3^{-7.5}\, x_{\rm H_2}^{-2}\, \zeta^{-0.29}\ {\rm cm^{-2}}, \tag{25}$$

$$\frac{\langle \dot M_{\rm TDE}\rangle}{\dot M_{\rm amb}} = 6.8\, f_{\rm B,-3}^{-1}\, M_{\rm h,6}^{2.2}\, T_3^{7.5}\, x_{\rm H_2}^{2}\, \zeta^{0.22}. \tag{26}$$

The steep temperature exponents in equations (23)–(26) arise because $n_{\rm bg} \propto \rho_{\rm MC}\sigma_{\rm cl}^2/v_{\rm c}^2$ and $\sigma_{\rm cl} \propto T_3^{-3.76}$; they are the reason that the temperature must be pinned down before any of these quantities can be quoted.

4.2. *The temperature is bracketed, not assumed*

Two natural guesses for $T_{\rm MC}$ both fail, and the way they fail is instructive.

The first is to set $T_{\rm MC}$ at the well-known floor of H_2-cooled gas, ~ 200 K, where the cooling rate collapses (Bromm et al. 2002; Abel et al. 2002). Evaluating equations (16)–(21) at $T_3 = 0.2$ gives clouds of radius 150 pc at Mach 1400 with $M_{\rm disk} \simeq 8\times10^{7}\, M_\odot$ for a $10^6\, M_\odot$ hole. Gas at 200 K radiates so little per unit mass in H_2 lines that the energy condition requires an enormous gas mass, and the clouds that it requires are two orders of magnitude larger than the disk that must contain them. This is not a solution.

The second is to treat $T_{\rm MC}$ as a fifth unknown closed by local thermal balance between turbulent dissipation and H_2 cooling, $\rho_{\rm MC}\sigma_{\rm cl}^3/2R_{\rm MC} = \Lambda_{\rm vol}$. The monomial system of five equations has a solution, $T_{\rm MC} = 420\, M_{\rm h,6}^{-0.36}$ K, but it too fails the geometry: $R_{\rm MC} = 9.4$ pc against $r_{\rm b} = 4.9$ pc, and the "clouds" outweigh the black hole. Turbulent dissipation alone cannot heat the gas enough. This is not a defect peculiar to the metal-poor case: in the GL26 solution itself the turbulent dissipation rate is only $\sim 5\times10^{-4}$ of the UDR power radiated by the clouds. The energy that the clouds radiate is deposited by the UDR shocks directly, as argued in GL26 (§2), and the turbulent cascade is a minor channel for it.

What does constrain $T_{\rm MC}$ is a pair of consistency requirements internal to the solution, and they act in opposite directions. The clouds must fit inside the disk that holds them; with $H = f_\Omega r_{\rm b}$ and equation (16),

$$\frac{H}{R_{\rm MC}} = 3.1\, M_{\rm h,6}^{1.51}\, T_3^{3.76}\, x_{\rm H_2} \ \geq 1$$
$$\Longrightarrow \quad T_{\rm MC} \geq 740\, M_{\rm h,6}^{-0.40}\, x_{\rm H_2}^{-0.27}\ {\rm K}. \tag{27}$$

This is the same requirement that GL26 use to bound f_* from below; here, because $R_{\rm MC} \propto T_3^{-3.76}$, it bounds the temperature from below instead. The clouds must also be supersonically turbulent. The engine works by injecting momentum that sustains the turbulent support that suppresses star formation, and that mechanism has no meaning for thermally supported clouds. Equation (17) gives

$$\mathcal{M} \geq 1$$
$$\Longrightarrow \quad T_{\rm MC} \leq 1100\, M_{\rm h,6}^{-0.36}\, x_{\rm H_2}^{-0.24}\ {\rm K}. \tag{28}$$

The two bounds bracket a window

$$740\, M_{\rm h,6}^{-0.40} \ \lesssim\ T_{\rm MC}\ \lesssim\ 1100\, M_{\rm h,6}^{-0.36}\ {\rm K}, \tag{29}$$

whose width, $T_{\rm max}/T_{\rm min} = 1.5\, M_{\rm h,6}^{0.05}$, is essentially independent of black hole mass, molecular fraction, and ζ. Figure 1 shows the belt quantities across the window for $M_{\rm h} = 10^6\, M_\odot$.

The window is narrow enough that the temperature is effectively determined, $T_{\rm MC} \simeq 900\, M_{\rm h,6}^{-0.38}$ K at its geometric center. Whether a single temperature can describe the clouds at all depends on how the energy reaches them, so we state it here. The cooling time of H_2 gas in the window is short, $t_{\rm cool} \simeq 8\, T_3^{-2.8} x_{\rm H_2}^{-1}$ yr, whereas a given cloud of radius $R_{\rm MC}$ is struck by a remnant only every $\tau_{\rm c}(R_{\rm UDR}/R_{\rm MC})^2 \simeq 5\times10^{3}$ yr. Were the energy delivered mechanically at the point of impact, the clouds would be cold almost everywhere and warm only in the $\sim 10\, M_\odot$ that each shell sweeps and dissociates (Hollenbach & McKee 1980), and the sound speed in equations (2) and (28) would be that of the cold bulk rather than of the emitting gas. That is not how the energy arrives. The remnants become radiative at $\sim 10^3\,{\rm km\,s^{-1}}$ (Section 3.3) and emit their 1.8×10^{50} erg as ∼keV X-rays over a few times $\tau_{\rm rad} \simeq 30$ yr. A new

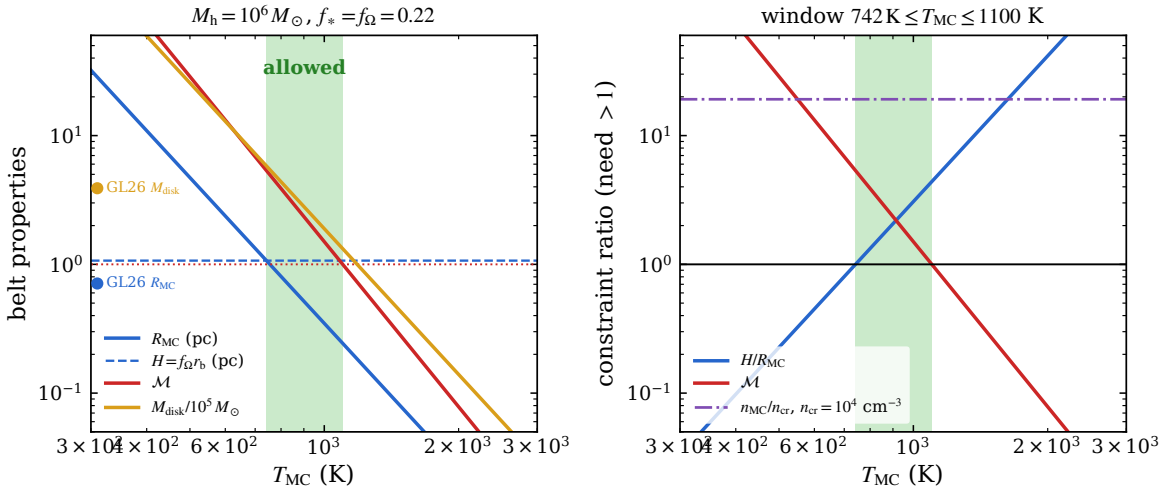


Figure 1. Left: cloud radius, Mach number and disk gas mass of the metal-poor belt as functions of the cloud temperature for $M_h = 10^6\,M_\odot$, $f_* = f_\Omega = 0.22$, $x_{H_2} = \zeta = 1$. The dashed line is the disk half-thickness $H = f_\Omega r_b$, which does not depend on temperature; the dotted line marks $\mathcal{M} = 1$. The corresponding GL26 values for the CO-cooled belt at 10 K are marked at the left edge for reference. Right: the two consistency ratios that must exceed unity, H/R_{MC} (clouds fit in the disk) and $\mathcal{M}$ (clouds are supersonically turbulent), together with the ratio of the cloud density to a nominal LTE critical density of 10^4 cm^{-3}. The shaded band is the allowed window of equation (29). The temperature of the H_2-emitting gas is not a free parameter of the model but is confined to a factor of 1.5.

remnant forms every $\Gamma_{eq}^{-1} \simeq 70$ yr, so at any moment roughly one is radiating somewhere in the belt. The photoabsorption cross-section of primordial gas at 1 keV, $\sim 2\times10^{-23}$ cm^2 per hydrogen nucleus, gives unit optical depth at $N_H \simeq 5\times10^{22}$ cm^{-2}, comparable to the column through the ambient medium (eq. 25) and a tenth of the column through a cloud. The X-rays are therefore absorbed throughout the cloud volumes rather than at their surfaces, and only the tail above ~ 3 keV escapes the belt. The clouds are therefore X-ray dissociation regions (Maloney et al. 1996; Meijerink & Spaans 2005; Wolfire et al. 2022). They are heated volumetrically and quasi-continuously by secondary electrons, which deposit a few tens of per cent of the absorbed energy as heat and the rest in ionization and excitation that is itself reradiated (Dalgarno et al. 1999). They cool through H_2 rotational lines at a few hundred to 10^3 K, which is where such models place dense irradiated gas and where equation (29) places it. Because $\Lambda_{vol} \propto T^{3.76}$, a fluctuation of a factor of three in the heating from one remnant to the next produces a fluctuation of only ~ 30 per cent in T_{MC}. A single temperature therefore describes the cloud gas to the accuracy of the calculation, and the sound speed in the virial and sonic conditions is that of the same gas that radiates. The X-rays that escape are a prediction, $L_X \lesssim 10^{40}$ erg s^{-1} in quiescence, orders of magnitude below the stacked limits on LRDs (Yue et al. 2024; Ananna et al. 2024; Maiolino et al. 2025; Kokubo & Harikane 2025). The X-ray weakness of the population therefore constrains the flares and not the engine.

At the center of the window and $M_h = 10^6\,M_\odot$, the belt has $R_{MC} = 0.52$ pc, $\mathcal{M} = 2.4$, $\sigma_{cl} = 4.3$ km s^{-1}, $M_{MC} = 3.6\times10^3\,M_\odot$, $N_{MC} = 77$, and $M_{disk} = 2.8\times10^5\,M_\odot$ (Table 1). Across the whole window M_{disk} ranges from 1.3×10^5 to $5.8\times10^5\,M_\odot$, straddling the $3.9\times10^5\,M_\odot$ of the metal-rich solution. It is at first surprising that a belt cooled by H_2 at $\sim 10^3$ K should so closely resemble one cooled by CO at 10 K, but the numbers explain it: at the belt density, the cooling per hydrogen nucleus of CO at 10 K, $\Lambda_0 n_{MC} \simeq 2.5\times10^{-22}$ erg s^{-1}, is within a factor of two of that of H_2 at 10^3 K, $L_{LTE}/2 \simeq 5\times10^{-22}$ erg s^{-1}. The two branches of the equilibrium therefore have nearly the same gross structure and differ in the thermodynamic state of the gas: a Mach number of ~ 30 in cold gas becomes ~ 2 in warm gas at the same turbulent velocity.

4.3. *The compression criterion and the mass floor*

The fourth condition of GL26 tests whether clouds with the derived properties can compress the cusp to the dispersion that the collision cap requires. Replacing equation (5) with $t_{relax,MC} = t_{relax,*}$ and re-solving the monomial system with the H_2 cooling function gives the largest dispersion the clouds can sustain,

$$\sigma_{comp} = 2.8\times10^2\,M_{h,6}^{0.43}\,T_3^{1.19}\,x_{H_2}^{0.32}\,\zeta^{0.05}\,\mathrm{km\,s^{-1}}, \quad (30)$$

to be compared with the GL26 value of $2.8\times10^2\,M_{h,6}^{0.40}$ km s^{-1} (272 km s^{-1} for the disk-fed geometry).

Table 1. The engine at $M_{\rm h} = 10^6\,M_\odot$, $f_* = f_\Omega = 0.22$

Quantity	Metal-rich (GL26)	Metal-poor (this work)
Coolant	CO, dust	H_2 rotational lines
$T_{\rm MC}$ (K)	10 (imposed)	740–1100 (derived)
$\sigma_{\rm eq}$ (km s^{-1})	240	240
$\Gamma_{\rm eq}$ (yr^{-1})	1.4×10^{-2}	1.4×10^{-2}
$a_{\rm h}$ (pc)	0.15	0.15
$r_{\rm b}$ (pc)	4.8	4.9
$n_{\rm MC}$ (cm^{-3})	1.9×10^5	1.9×10^5
$R_{\rm MC}$ (pc)	0.71	0.52
$\mathcal{M}$	32	2.4
$\sigma_{\rm cl}$ (km s^{-1})	6.0	4.3
$M_{\rm disk}$ ($M_\odot$)	3.9×10^5	2.8×10^5
$N_{\rm MC}$	40	77
$L_{\rm MC}$ (erg s^{-1})	8×10^{40} (CO)	8×10^{40} (H_2)
$n_{\rm bg}$ (cm^{-3})	7.4×10^3	3.9×10^3
$N_{\rm H}$ (cm^{-2})	1.1×10^{23}	5.9×10^{22}
A_V	50	$27\,Z'$
$L_{\rm AGN}/L_{\rm Edd}$ ($f_{\rm B,-3}$)	0.051	0.026
$\langle\dot M_{\rm TDE}\rangle/\dot M_{\rm amb}$ ($f_{\rm B,-3}^{-1}$)	1.6	3.1
$\dot p_{\rm TDE}/\dot p_{\rm AGN}$	9.4	$\gtrsim 30$
Mass floor ($M_\odot$)	7.3×10^5	$7.0\times10^5\,T_3^{-3.0}$
$t_{\rm gas}$ (yr)	5.5×10^7	4.0×10^7

Note—Metal-poor values are evaluated at the center of the temperature window, $T_{\rm MC} = 900$ K, with $x_{\rm H_2} = \zeta = 1$. Quantities in the upper block are set by the cusp or by the covering condition and do not depend on composition.

Requiring $\sigma_{\rm comp} \geq \sigma_{\rm eq}$ gives the mass floor

$$M_{\rm h} \geq 7.0\times10^5\,T_3^{-3.0}\,x_{\rm H_2}^{-0.80}\,\zeta^{-0.12}\;M_\odot, \tag{31}$$

which is $7\times10^5\,M_\odot$ at $T_3 = 1$ and $1.4\times10^6\,M_\odot$ at $T_3 = 0.8$. The metal-poor engine, like the metal-rich one, cannot run around the $\sim 10^4\,M_\odot$ seeds that runaway collisions produce (Pacucci et al. 2025; Inayoshi et al. 2020). It switches on only after the seed has grown by two orders of magnitude, and the argument of GL26 (§6.1) about the resulting delay carries over. We note that the T_3^{-3} dependence pulls in the same direction as the window of equation (29): holes near the floor have $T_{\rm min} \simeq 850$ K and a floor of $\sim 10^6\,M_\odot$, so the floor for the metal-poor engine is best quoted as $M_{\rm h} \gtrsim 10^6\,M_\odot$.

4.4. *The flattening window revisited*

GL26 found that three requirements bound the cusp flattening to $0.18 \lesssim f_* \lesssim 0.25$: that TDE momentum outweigh AGN radiation pressure ($f_* \gtrsim 0.12$), that the clouds fit within the disk ($f_* \gtrsim 0.18$), and that the cusp be no denser than observed nuclear star clusters ($f_* \lesssim 0.25$ for $\rho_0 \leq 10^8\,M_\odot\,{\rm pc}^{-3}$). Two of the three change character at low metallicity.

The AGN bound disappears. In GL26 the radiation pressure couples to the gas through dust, $\dot p_{\rm rad} = \tau L_{\rm AGN}/c$ with $\tau \simeq 30\,f_{\rm Edd}$ for solar dust (Li & Draine 2001; Nenkova et al. 2008; Netzer 2006). In metal-poor gas $\tau \to \max(30 f_{\rm Edd} Z',\ \tau_{\rm es})$, where the electron-scattering optical depth through the belt is $\tau_{\rm es} = N_{\rm H}\sigma_{\rm T} \simeq 0.04$. With $f_{\rm Edd}$ from equation (24) and $\dot p_{\rm TDE} = 1.7\times10^{33}(\Gamma_{\rm eq}/10^{-2}\,{\rm yr}^{-1})$ g cm s^{-2} from GL26's equations (12), (17) and (19), the ratio at $f_* = 0.22$ is $\dot p_{\rm TDE}/\dot p_{\rm AGN} \simeq 30$ for $Z' = 1$ and $\simeq 300$ for $Z' = 0.1$. Tidal disruptions dominate the feedback budget for any flattening the other constraints allow.

The geometric bound is replaced by the pair of temperature bounds. Because $R_{\rm MC}$ now depends on $T_{\rm MC}$, the requirement $H \geq R_{\rm MC}$ is a lower bound on $T_{\rm MC}$ at each f_* (eq. 27) rather than a bound on f_* alone, and the sonic requirement $\mathcal{M} \geq 1$ supplies the upper bound

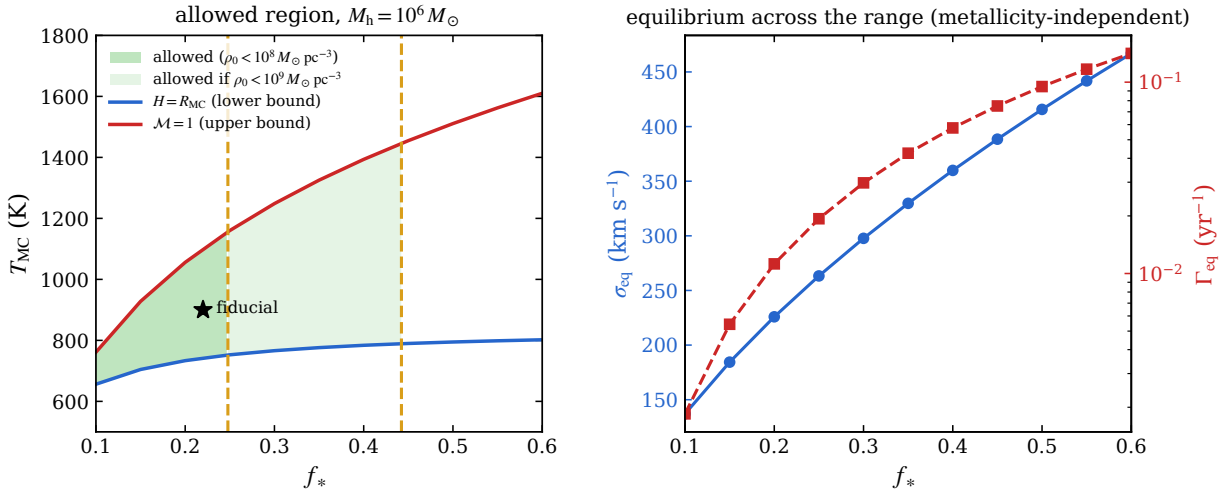


Figure 2. Left: the allowed region for the metal-poor engine in the plane of cusp flattening f_* (with $f_\Omega = f_*$, $b = 1$) and cloud temperature, for $M_h = 10^6\,M_\odot$. The lower curve is the geometric requirement $H = R_{MC}$, the upper curve the sonic requirement $\mathcal{M} = 1$. The vertical dashed lines mark the flattening at which the central density of the cusp reaches 10^8 and $10^9\,M_\odot\,\mathrm{pc}^{-3}$; the darker shading is allowed under the GL26 ceiling and the lighter shading under the relaxed ceiling appropriate to LRD cores. The bound of GL26 from AGN dominance does not appear because TDE momentum exceeds AGN radiation pressure everywhere on the plotted range at low metallicity. The star marks the fiducial solution. Right: the equilibrium dispersion and disruption rate across the same range of f_*; both are set by the cusp alone and are independent of metallicity.

(eq. 28). Repeating the solution for the collision cap and the belt across $0.10 \le f_* \le 0.60$ gives the allowed region in the (f_*, T_{MC}) plane shown in Figure 2. It is a wedge that widens toward rounder cusps: at $f_* = 0.10$ the window is 660–760 K, at 0.22 it is 740–1100 K, and at 0.40 it is 780–1400 K. The density ceiling is unchanged, since ρ_0 depends only on the cusp: $f_* \le 0.25\,M_{h,6}^{0.45}$ for $\rho_0 \le 10^8\,M_\odot\,\mathrm{pc}^{-3}$, or $f_* \le 0.44\,M_{h,6}^{0.45}$ if the ceiling is relaxed to the $\sim 10^9\,M_\odot\,\mathrm{pc}^{-3}$ inferred for the most extreme LRD cores (Pacucci et al. 2025). The fiducial $f_* = 0.22$ sits inside the allowed region for either choice.

4.5. *The mass range of the metal-poor engine*

Three considerations confine the metal-poor engine to a narrow range of black hole mass. The floor of equation (31) is $\sim 10^6\,M_\odot$. The LTE treatment of the H_2 cooling requires $n_{MC} \gtrsim n_{cr}$, where $n_{cr} \sim 10^4$ cm^{-3} for collisions with H and $\sim 10^5$ cm^{-3} for collisions with H_2 (Section 3.2). With $n_{MC} \propto M_{h,6}^{-2}$ (eq. 12) this holds for $M_{h,6} \lesssim 1.5$–5, depending on the collider. At higher masses the cooling per molecule is suppressed by $(1 + n_{cr}/n_{MC})^{-1}$, which acts like a reduced x_{H_2} and pushes the temperature window (eqs. 27–28) upward. And the gravitational focusing that sets the collision rate requires $\sigma_h < v_{esc,*}$, which GL26 show fails above $\sim 10^8\,M_\odot$. The binding constraints are the first two, so the metal-poor engine operates for

$$10^6 \lesssim M_h \lesssim (1.5\text{–}5) \times 10^6\,M_\odot, \qquad (32)$$

with disruption rates of $\Gamma_{eq} \simeq (0.5\text{–}2) \times 10^{-2}$ yr^{-1} and cloud temperatures of 400–1100 K across the range. This is the low end of the 10^6–$10^9\,M_\odot$ black hole masses inferred for LRDs from virial estimates on their broad Balmer lines (Harikane et al. 2023; Matthee et al. 2024; Greene et al. 2024). Those estimates may be biased high: if the line wings are produced by electron scattering rather than Doppler motion, as the highest-quality spectra suggest, the intrinsic line cores imply masses of 10^5–$10^7\,M_\odot$ (Rusakov et al. 2026), which brackets the engine's range.

5. CONSEQUENCES FOR LITTLE RED DOTS

5.1. *Obscuration*

The most robust prediction of the metal-rich engine was $A_V \simeq 50$ through the gas, independent of geometry, and GL26 identified it as the principal obstacle to an LRD application. It falls away at low metallicity for two reasons. The column itself is lower, $N_H \simeq 6 \times 10^{22}$ cm^{-2} at the window center (eq. 25, Table 1), because warm clouds at Mach 2 exert less pressure on the ambient medium than cold clouds at Mach 30 and the ambient density is correspondingly lower. And the dust-to-gas ratio scales with Z'. Together,

$$A_V \simeq 12\,Z'\,M_{h,6}^{-4.0}\,T_3^{-7.5}\,x_{H_2}^{-2}\,\zeta^{-0.29} \simeq 27\,Z' \qquad (T_3 = 0.9,\ M_{h,6} = 1), \qquad (33)$$

so that a nucleus at $Z' = 0.1$ is seen through $A_V \simeq 2.7$ even along sightlines through the disk, and at $Z' =$

0.01 is effectively unobscured. The reprocessed infrared emission is small for a different reason, one that does not scale with Z' once $A_V \gtrsim 1$. The belt intercepts a fraction $\sim f_\Omega$ of the sky seen from the hole and absorbs what it intercepts whenever $A_V \gtrsim 1$. It is 30 light-years across, however, so its dust responds not to the flare being observed but to the luminosity averaged over the duty cycle of Section 5.2, $\langle L \rangle \simeq F_{\rm Edd} L_{\rm Edd} + L_{\rm AGN} \simeq 8 \times 10^{42}\, M_{\rm h,6}$ erg s^{-1}. Dust at $r_{\rm b}$ then sits at $T_{\rm d} \simeq 60$ K and radiates $L_{\rm IR} \lesssim f_\Omega \langle L \rangle \simeq 2 \times 10^{42}$ erg s^{-1} at rest-frame ~ 50 μm, about one per cent of the flare luminosity. This is far below the hot-dust limits from MIRI, which probe dust at $\gtrsim 300$ K within a fraction of a parsec (Setton et al. 2025; Chen et al. 2025). The $\lesssim 3 \times 10^3\, Z'\, M_\odot$ of dust in the belt is also three orders of magnitude below the cold-dust limits from ALMA (Casey et al. 2025). The dust deficit of LRDs is therefore not in tension with a metal-poor engine; it is what a metal-poor engine predicts.

The gas column is a different matter. The belt has $N_{\rm H} \sim 10^{23}$ cm^{-2} at ~ 5 pc, in gas at $n \sim 10^3$–10^5 cm^{-3} and $\sim 10^3$ K. The "black hole star" interpretation instead requires $N_{\rm H} \gtrsim 10^{24}$ cm^{-2} of much denser gas at $\sim 10^3$ gravitational radii to produce a Balmer break by gas absorption (Naidu et al. 2026). The engine does not itself construct that envelope. It is not implausible, however, that one forms in a nucleus in which a $10^6\, M_\odot$ hole is fed intermittently at super-Eddington rates by disruptions, with $\sim 3 \times 10^5\, M_\odot$ of gas within a few parsecs.

5.2. *Luminosity and duty cycle*

The engine does not power an LRD continuously. The Bondi-fed luminosity of equation (24) is $\sim 3 \times 10^{-2} f_{\rm B,-3} L_{\rm Edd} \simeq 3 \times 10^{42}\, M_{\rm h,6}$ erg s^{-1}, two orders of magnitude below the bolometric luminosities of 10^{44}–10^{45} erg s^{-1} inferred for LRDs. What the engine supplies is a high rate of flares. Following GL26's equation (13), the fraction of time that the hole spends above its Eddington luminosity from disruption fallback is $F_{\rm Edd} = 1.3 \times 10^{-2}\, M_{\rm h,6}^{-0.4}\, \Gamma_{-2} \simeq 1.9 \times 10^{-2}\, M_{\rm h,6}^{-1.24}$. The fraction above $0.1\, L_{\rm Edd}$ is about four times larger for a $t^{-5/3}$ decay, $F_{0.1} \simeq 7 \times 10^{-2}\, M_{\rm h,6}^{-1.24}$. This holds provided that the peak rate of fallback exceeds Eddington, as it does for $M_{\rm h} \lesssim 10^7\, M_\odot$, and that the duration is set by the decaying tail rather than the rise. An engine nucleus is therefore observed as a bright, broad-lined source ~ 2–7 per cent of the time, at luminosities near and above $L_{\rm Edd} \simeq 1.3 \times 10^{44}\, M_{\rm h,6}$ erg s^{-1}.

This bears directly on the abundance argument. Bellovary (2025) noted that a rate of $\sim 10^{-4}$ yr^{-1} per system is needed to reconcile the observed LRD density with the predicted abundance of clusters that undergo runaway collapse. This rate is equivalent to a duty cycle of $\sim 10^{-4}$ for year-long flares, and they noted that it is in tension with the halo mass function unless the initial mass function is top-heavy. With a duty cycle of $\sim 7 \times 10^{-2}$ instead, the required density of engine hosts is

$$n_{\rm host} \simeq \frac{n_{\rm LRD}}{F_{0.1}} \simeq \frac{10^{-5}\ {\rm cMpc^{-3}}}{7 \times 10^{-2}} \simeq 1.4 \times 10^{-4}\ {\rm cMpc^{-3}}, \tag{34}$$

using the spectroscopic number density of LRDs from Matthee et al. (2024). The photometric censuses give values up to an order of magnitude higher (Kokorev et al. 2024; Akins et al. 2025), and $n_{\rm host}$ scales with them. This is nearly three orders of magnitude fewer hosts than the low-rate picture requires, and comparable to the abundance of halos of a few $\times 10^{11}\, M_\odot$ at $z \sim 5$ (Sheth & Tormen 1999), which relaxes the tension without appeal to the initial mass function. GL26 made the same point qualitatively. With the metal-poor solution we can make it quantitatively, for the environment in question.

The same duty cycle leads to a prediction about variability. Above $L_{\rm Edd}$ the luminosity is capped near Eddington while the fallback rate is not, so the brightest ~ 2 per cent of the time is a plateau with little variation. During the remaining ~ 5 per cent, between 0.1 and 1 $L_{\rm Edd}$, the luminosity declines as $t^{-5/3}$ over a few years. A rest-frame baseline of $\Delta t \simeq 0.3$–0.5 yr samples a change of $(5/3)\Delta t/t \simeq 20$–$40$ per cent at $t \simeq 2$ yr, and such baselines are typical of multi-epoch imaging with JWST at $z \sim 5$. Existing studies find no variability in most LRDs, at limits of a few tenths of a magnitude, and a minority of strongly variable candidates (Kokubo & Harikane 2025; Tee et al. 2025; Zhang et al. 2025). Simulated light curves show that super-Eddington disks reproduce that absence where sub-Eddington models of AGN variability do not, and that the ongoing high-cadence campaigns with JWST will detect continuum changes of $\Delta m > 1$ only for the lowest-mass sources (Secunda et al. 2026). The Eddington-capped plateau of the engine sits on the same side of that comparison. Roughly a quarter of engine LRDs would be caught on the plateau and the rest in the tail, at the level that those limits are beginning to probe. The population statistics are therefore a marginal test now and will be a sharp one with longer baselines. Two effects would soften the tail further: the electron-scattering envelope invoked to explain the line wings (Rusakov et al. 2026) smooths and delays the variations, and the $2M_{\rm h}$ of stars inside $a_{\rm h}$ flattens the fallback below $t^{-5/3}$ at late times (Oña et al. 2025). What is firm is the repetition: an engine

nucleus produces a fresh flare every $\Gamma_{\rm eq}^{-1} \simeq 70\, M_{\rm h,6}^{0.84}$ yr, so an LRD that has faded should rebrighten within a human lifetime.

5.3. *Black hole growth*

Equation (26) gives $\langle \dot{M}_{\rm TDE} \rangle / \dot{M}_{\rm amb} \simeq 3\, f_{\rm B,-3}^{-1}$ at the window center, compared with 1.6 in the metal-rich disk-fed solution. The lower ambient density of the warm belt tips the balance toward the disrupted stars, and for any Bondi efficiency below 3×10^{-3} the hole grows predominantly in the "black tide" mode of Young et al. (1977). N-body models of dense nuclear clusters around 10^3–$10^5\, M_\odot$ holes find that feeding by stars dominates there as well (Kritos & Silk 2026b; Wang et al. 2026b). Over the gas-limited lifetime $t_{\rm gas} = M_{\rm disk}/m_* \Gamma_{\rm eq} = 4 \times 10^7\, M_{\rm h,6}^{0.33}\, T_3^{-3.76} x_{\rm H_2}^{-1}$ yr the hole consumes $\sim 2 \times 10^5\, M_\odot$ in stellar material, of order 20 per cent of its mass, in discrete super-Eddington episodes. If the disk is replenished from the host, the ceiling is the cusp consumption time $t_{\rm cusp} = N_*/\Gamma_{\rm eq} = 2.8 \times 10^8\, M_{\rm h,6}^{1.84}$ yr, over which the hole roughly doubles.

5.4. *Warm molecular hydrogen*

The energy condition (4) is a statement about where the injected power goes, and in the metal-poor engine it goes into H_2 rotational lines: $L_{\rm H_2} \simeq 8 \times 10^{40}\, M_{\rm h,6}^{-0.5}$ erg s^{-1} from $\sim 3 \times 10^5\, M_\odot$ of gas at $\sim 10^3$ K (eq. 22). At these temperatures and densities the emission is carried by the pure rotational 0–0 S(0)–S(5) lines at 28.2, 17.0, 12.3, 9.7, 8.0 and 6.9 μm, with a subdominant contribution from the 1–0 rovibrational lines near 2 μm. Warm H_2 radiating dissipated mechanical energy at this level is a known phenomenon: the H_2-luminous radio galaxies and the intergalactic shock in Stephan's Quintet have $L_{\rm H_2} \sim 10^{40}$–10^{42} erg s^{-1} in the rotational lines, excitation temperatures of a few hundred to 10^3 K, and H_2 emission that exceeds the X-ray and dust emission of the same gas (Appleton et al. 2006; Ogle et al. 2010; Guillard et al. 2012; Lanz et al. 2015). The engine belt is a parsec-scale, metal-poor version of the same thing.

No such emission has been reported from an LRD, and we are aware of no published search for it. The only H_2 emission so far reported at $z \gtrsim 7$ is tentative far-ultraviolet fluorescence in stacked spectra of ordinary galaxies (Johnson et al. 2025), a different mechanism and gas phase. The reasons are instrumental rather than physical. At $z \sim 5$ the rotational lines redshift to 40–170 μm, beyond the reach of JWST. The 1–0 S(1) line at 2.12 μm falls at ~ 13 μm, within the range of the MIRI medium-resolution spectrograph. If the vibrational lines carry tens of per cent of $L_{\rm H_2}$, its predicted flux is a few $\times 10^{-20}$ erg s^{-1} cm^{-2}, an order of magnitude below what MIRI spectroscopy has achieved on the same line in an AGN host at cosmic noon (Kakkad et al. 2025). The H_2 luminosity is moreover three to four orders of magnitude below the flare luminosity of the LRD itself. The absence of a detection at high redshift therefore carries no information about the model at present.

What the LRD literature does contain is evidence for cool molecular gas of a different kind. Two LRDs at $z \simeq 2$ show water-vapour absorption bands at rest-frame ~ 1.4 μm, implying a photosphere-like envelope at $T \sim 2000$–4000 K (Wang et al. 2026a), and LRDs identified in the local Universe show Na D, K I, Ca II and other low-ionization absorption features characteristic of cool gas envelopes (Lin et al. 2026). These are absorption signatures of dense gas at $\gtrsim 10^8$ cm^{-3} within light-days of the black hole, the envelope invoked by the "black hole star" interpretation (Naidu et al. 2026), not emission from parsec-scale clouds at 10^5 cm^{-3}, and they do not bear on the belt directly. They do establish that molecular gas is present in LRDs and that their nuclei are cooler and denser than those of ordinary broad-line AGN, which is the environment that the engine constructs at larger radii.

The prediction is therefore best tested elsewhere. For the local LRDs of Lin et al. (2026), at $z \sim 0.1$, the rotational S(1)–S(5) lines and the rovibrational lines fall within reach of MIRI spectroscopy, at fluxes $\sim 10^5$ times those of their high-redshift counterparts. MIRI has already measured seven H_2 rotational lines from gas at $n \sim 10^5$ cm^{-3} in I Zw 18, at 3 per cent of solar metallicity (Hunt et al. 2025). Any such object with a black hole of $\sim 10^6\, M_\odot$ and a recent starburst is the natural place to look for a compact ($\lesssim 10$ pc) reservoir of a few $\times 10^5\, M_\odot$ of warm molecular hydrogen at $\sim 10^3$ K, at Mach 2–4, and without the cold CO-emitting phase that dominates metal-rich nuclei. A detection would fix $T_{\rm MC}$ observationally through the excitation temperature of the rotational lines, and equation (29) gives the value to expect.

5.5. *The molecular fraction*

The solution requires the belt gas to be predominantly molecular, since $R_{\rm MC} \propto x_{\rm H_2}^{-1}$ and the temperature window shifts as $x_{\rm H_2}^{-1/4}$ (eqs. 27–28). In truly primordial gas, without dust, H_2 forms through the H^- channel and reaches fractions of only $x_{\rm H_2} \sim 10^{-3}$–10^{-2} even behind ionizing shocks (Shapiro & Kang 1987), since the electrons that catalyse it recombine faster than the molecules form. Only above $n \gtrsim 10^8$ cm^{-3} do three-body reactions convert the gas fully (Palla et al. 1983). At $x_{\rm H_2} = 10^{-2}$ the window of equation (29) moves to

2500–3300 K, where H_2 is collisionally dissociated at $n \sim 10^5$ cm^{-3}, and the H_2 branch does not exist. The belt is not unirradiated primordial gas, however: the X-ray heating of Section 4.2 maintains an electron fraction of a few $\times 10^{-3}$, and the H^- channel then forms H_2 on a timescale of $\sim 5 \times 10^4$ yr regardless of Z'. We do not attempt the calculation of XDR chemistry that would show whether that channel can drive the gas fully molecular against X-ray dissociation, so we do not rely on it.

The situation is different for even trace metallicity. Grain-catalysed formation proceeds at $\sim 3 \times 10^{-17} Z'$ cm^3 s^{-1} per hydrogen nucleus in cold gas (Hollenbach & McKee 1979), reduced by a factor of ~ 10 at the 10^3 K gas temperature of the belt because the sticking and formation efficiencies fall with gas temperature (Hollenbach & McKee 1979; Cazaux & Tielens 2004). It converts gas at $n_{\rm MC} = 2 \times 10^5$ cm^{-3} to H_2 on a timescale $t_{\rm form} \simeq 5 \times 10^4\, Z'^{-1}$ yr, shorter than the crossing time (eq. 13) for $Z' \gtrsim 0.2$ and shorter than the gas-limited lifetime of the engine, $t_{\rm gas}$ (Section 5.3) for $Z' \gtrsim 10^{-3}$. Each UDR dissociates only the $\sim 10\, M_\odot$ that it sweeps, a small fraction of a $\sim 10^3\, M_\odot$ cloud, and the Lyman–Werner radiation from the nucleus is self-shielded at the $N_{H_2} \gtrsim 10^{22}$ cm^{-2} columns of the clouds (Draine & Bertoldi 1996; Wolcott-Green et al. 2011). Gas that arrives in the belt molecular therefore stays molecular, and gas that arrives atomic becomes molecular within the lifetime of the belt for $Z' \gtrsim 10^{-3}$. At the other end, the metal fine-structure lines reach ~ 15 per cent of the H_2 cooling at $Z' \simeq 0.3$ (Section 3.2), and by $Z' \simeq 0.7$ the cold CO branch fits within the disk again (Section 3.1). Between the two, the belt would settle at a few hundred K with mixed cooling, an intermediate state that we have not solved. We therefore regard $x_{H_2} \simeq 1$ and pure H_2 cooling as appropriate for $10^{-3} \lesssim Z' \lesssim 0.3$. This is the metal-poor engine's regime of validity. It brackets the metallicities at which runaway collisions are expected to build the LRD seeds (Pacucci et al. 2025; Inayoshi et al. 2020), and it contains the measured metallicities of LRD hosts, $Z' \simeq 0.08$ with a spread of 0.6 dex and two objects below 0.013 (Nikopoulos et al. 2026). Below $Z' \sim 10^{-3}$ the belt would have to be cooled by atomic processes, through Lyα at ~ 8000 K, with $\Lambda_{\rm vol} \propto n^2$ and a dependence on the ionization fraction that the UDR shocks themselves would set. That branch, if it exists, would need a separate treatment.

6. CAVEATS

The treatment inherits every simplification of GL26, and §6.5 of that paper applies without change: the loss-cone rate sits near $q \sim 1$ and requires the bridged flux; the normalization of $\Gamma_{\rm eq}$ is uncertain by two orders of magnitude through the collision coefficient $\Lambda_{\rm c}$ while its mass exponent is stable; the compression criterion rests on a plausibility argument rather than a computed torque; and the clouds are treated as a single population at a single radius. To these we add the following.

The cooling function of equation (7) is the rotational LTE rate alone. Including the sub-thermal vibrational contribution at $n \sim 10^5$ cm^{-3} would raise the cooling per molecule by tens of per cent and steepen its temperature dependence somewhat, shrinking $R_{\rm MC}$ and $M_{\rm disk}$ by the same factor. Both effects are smaller than the factor of two to which the calculation is accurate, and they are absorbed into x_{H_2}.

The temperature window is bounded by two conditions of different character. The lower bound, $H \geq R_{\rm MC}$, is the same geometric requirement that GL26 use and is secure. The upper bound, $\mathcal{M} \geq 1$, marks where the framework ceases to apply rather than where a solution ceases to exist: a belt of thermally supported clouds at ~ 1500 K is a self-consistent configuration, but not one in which UDR momentum sustains turbulent support against star formation, which is the mechanism on which the engine rests. A treatment of the transition from turbulent to thermal support would be needed to say what happens at the upper edge.

We have assumed that the belt gas is fully molecular and argued that dust-catalysed formation makes this so for $Z' \gtrsim 10^{-3}$. The argument uses the rate of H_2 formation on dust grains in the solar neighbourhood, scaled linearly with Z'; if the dust-to-gas ratio falls faster than linearly at low metallicity, as observations of dwarf galaxies below $Z' \sim 0.1$ suggest (Rémy-Ruyer et al. 2014; Galliano et al. 2018), the lower edge of the validity range moves up accordingly.

Finally, the energy budget of equation (4) assigns the full UDR luminosity to the clouds, as GL26 do. In the picture of Section 4.2 that energy arrives as X-rays, and whatever escapes the belt above ~ 3 keV, or is deposited in the ambient medium rather than in the clouds, is lost to the budget. We estimate the loss at tens of per cent, comparable to the other uncertainties absorbed into x_{H_2}. The escaping part is a prediction rather than a defect. The natural next step is a calculation of the X-ray transfer and XDR chemistry of the belt, which would fix the heating efficiency, the escaping spectrum and the molecular fraction together.

7. SUMMARY

We have re-derived the self-sustaining tidal disruption engine of Guillochon & Loeb (2026) with the cooling

function appropriate to metal-poor gas, in which the CO and dust that hold the metal-rich belt at 10 K are absent and the clouds cool instead through H_2 rotational lines in LTE. Our conclusions are these.

1. The equilibrium disruption rate, stellar dispersion, cusp density, belt radius and cloud density are independent of composition, because the collision cap involves only the cusp and the covering condition fixes the belt radius without reference to the cooling. $\Gamma_{\rm eq} \simeq 1.4 \times 10^{-2}\, M_{\rm h,6}^{-0.84}$ yr^{-1}, $r_{\rm b} \simeq 4.9\, M_{\rm h,6}$ pc and $n_{\rm MC} \simeq 1.9 \times 10^5\, M_{\rm h,6}^{-2}$ cm^{-3} carry over from GL26 unchanged.

2. The cold branch of the equilibrium does not exist at low metallicity: with $\Lambda_0 \propto Z'$ the clouds outgrow the disk for $Z' \lesssim 0.7$, and the 10 K thermostat is gone in any case. The engine relocates to a warm branch, in which the clouds are heated throughout their volume by the keV emission of the radiative remnants, one of which is radiating in the belt at any moment. On this branch the temperature of the H_2-emitting gas is not imposed. It is bracketed to $740\, M_{\rm h,6}^{-0.40} \lesssim T_{\rm MC} \lesssim 1100\, M_{\rm h,6}^{-0.36}$ K by the requirements that the clouds fit within the disk and remain supersonically turbulent.

3. Within that window the warm belt has nearly the same cloud radii, gas mass ($\sim 3 \times 10^5\, M_\odot$) and mass floor ($\sim 10^6\, M_\odot$) as the metal-rich belt, because H_2 at 10^3 K and CO at 10 K happen to radiate comparably per hydrogen nucleus at the belt density. What differs is the state of the gas: Mach 2 rather than Mach 30.

4. The observable consequences differ sharply. The extinction is $A_V \simeq 27\, Z'$ rather than 50. The belt dust is heated only by the luminosity averaged over the duty cycle, so it reradiates ~ 1 per cent of the flare luminosity at ~ 60 K, and the dust deficit of LRDs is expected rather than problematic. The flares recur every ~ 70 yr and, outside their Eddington-capped plateau, decline by tens of per cent over rest-frame half-year baselines. AGN radiation pressure is negligible, so the lower bound on the cusp flattening moves from feedback dominance to the sonic condition, and the allowed region in the $(f_*, T_{\rm MC})$ plane is a wedge that widens toward rounder cusps. The black hole grows predominantly through disruptions. The injected energy emerges as warm H_2 emission at $L_{\rm H_2} \simeq 8 \times 10^{40}\, M_{\rm h,6}^{-0.5}$ erg s^{-1}.

5. The metal-poor engine operates for $10^6 \lesssim M_{\rm h} \lesssim (1.5\text{–}5) \times 10^6\, M_\odot$ and $10^{-3} \lesssim Z' \lesssim 0.3$. The mass range is bounded below by the compression criterion and above by the LTE requirement on the H_2 cooling. The metallicity range is bounded below by the need for dust-catalysed H_2 formation within the belt lifetime and above by the point at which cooling by metal fine-structure lines becomes competitive, and the cold CO branch returns at $Z' \gtrsim 0.7$. This range contains the measured metallicities of LRD hosts. Because the engine is above a tenth of the Eddington luminosity ~ 7 per cent of the time, it requires an LRD host abundance of $\sim 10^{-4}$ cMpc^{-3}, nearly three orders of magnitude fewer hosts than a 10^{-4} yr^{-1} rate would need.

In the picture that emerges, a nucleus has just finished building a $\sim 10^4\, M_\odot$ seed by runaway collisions in metal-poor gas and grows it by accretion to $\sim 10^6\, M_\odot$. It then enters a collisionally regulated state in which it disrupts a star every century and is seen as an LRD a few per cent of the time. It is surrounded not by the cold, dusty, CO-bright molecular disk of a metal-rich engine but by a few $\times 10^5\, M_\odot$ of warm, dust-poor molecular hydrogen at a thousand kelvin.

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The manuscript source, the bibliography, and the equilibrium-solver and figure scripts are available online.[3] The authors take full responsibility for the contents of this paper. Large language models (Claude Fable 5.1, Grok 4.6, and GPT5.6 Sol) were used, under author supervision, to draft and edit the text, to derive and check analytic results, to write and refine the analysis and figure-generation scripts released with this paper, and to refine the presentation of the figures. The figures themselves are produced by those scripts from the solver outputs; no generative model created or altered the plotted data. Every displayed equation has been rederived independently, every entry in the bibliography was validated against the NASA Astrophysics Data System, and all references and AI edits were hand-checked by the authors. We aimed to be compliant with the AAS guidelines on generative AI in manuscript preparation.[4] Those guidelines do not treat such tools as authors; responsibility for accuracy remains with the listed authors. We thank J. Xavier Prochaska for providing access to

[3] https://github.com/guillochon/engine-lrds
[4] https://journals.aas.org/author-llm-guidelines

AI models that were used to conduct independent AI reviews.

Software: `numpy`, `scipy`, `sympy`, `matplotlib`; the `tde-engine` scripts of GL26.

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